PATIENT SAFETY ESSENTIALS TOOLKIT: 5 Whys: Finding the Root Cause of a Problem

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Template: 5 Whys

EVENT. What happened? Define the problem as an *event*:

Michael Taylor, a 25-year-old patient at Virginia Hospital Center (VHC), suffered brain damage and permanent physical impairments following a laparoscopic cholecystectomy (gallbladder removal). His surgeon, Dr. Booth, entered a post-operative order for continuous pulse oximetry into the Cerner EHR system. Due to software design defects, the order was not executed as intended, and Taylor's oxygen levels were not continuously monitored, causing an undetected drop in blood oxygen levels that resulted in severe brain damage.

PATTERN. What's been happening? Define the problem as a *pattern* by selecting a poor performance factor:

EHR software systems with design defects that silently override physician orders or misrepresent order parameters lead to medication and care administration errors, patient harm, and legal liability — representing a recurring failure of EHR systems to communicate order changes transparently to clinicians and nursing staff.

STRUCTURE. Why is it happening? What are the tangible and intangible structures determining the results we see?

1. Taylor did not receive continuous pulse oximetry as ordered, and his dangerously low oxygen level went undetected during recovery.

Why is that?

2. The Cerner EHR system silently overrode Dr. Booth's intended immediate start time to a default 10:00 a.m. pharmacy start time, without any alert or notification to the physician or nursing staff.

Why is that?

3. Cerner's order entry screen was hardcoded to display the original entry time even after a hidden override occurred; the software contained a design defect that misrepresented the effective start time to the ordering physician.

Why is that?

4. The EHR system also presented 30 pending orders in a bulk unsigned-order list that could be approved all at once, burying the pulse oximetry order at the bottom without requiring individual review; this second design defect enabled Dr. Booth to transmit orders without seeing the pulse oximetry entry.

Why is that?

5. (Root Cause) Cerner did not comply with HIMSS industry standards requiring EHR systems to alert users when order parameters are changed and failed to warn hospital staff and physicians about these known design defects — leaving clinicians unaware that the system could silently modify their orders.

ACTION. What are the implications for action? What can you do to change the results?

- Require EHR vendors to display an explicit real-time alert whenever the system overrides or modifies a physician's entered order parameters (e.g., start time, route, frequency), ensuring the prescribing clinician sees and acknowledges the change before the order is transmitted.
- Redesign bulk order-approval workflows to require individual review and sign-off for high-risk or time-sensitive orders (such as monitoring and continuous care orders), preventing critical orders from being buried in batch approval queues.
- Mandate EHR vendors to proactively disclose known software design limitations to hospital customers and clinical end-users,

and provide formal training on how the system handles order entry, scheduling logic, and any default overrides.
• Establish interprofessional post-operative monitoring protocols at VHC and similar hospitals that require nursing staff to independently verify all time-sensitive monitoring orders against the original physician intent — especially following system implementations or upgrades — rather than relying solely on EHR-generated task prompts.